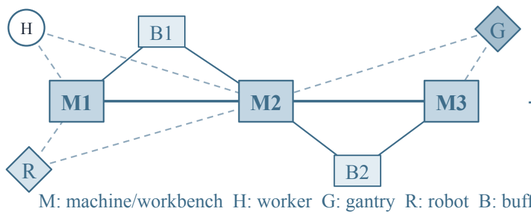
PHGST-DP

(a) Heterogeneous resource representation



Graph G
M^G, E_lap

Node-aligned history

6 groups

Grouped Linear + GELU
Mean aggregation

X: 30 × 38 × 27

Time + Laplacian
+ node embeddings

(b) Process-informed supervision and pattern clustering

Operational evidence

Queue | Stall / shortage | Logistics

Resource-specific rules + persistence

Raw

Sustained

Remove short runs, then fill gaps

Event / state targets

Cause targets

training supervision

History-derived conditioning

Rule-based regime IDs

Regime embedding

Local + neighbour cues

Precursor MLP

Short-pattern clustering

Activity patches
(length 3)

k-means

16 prototypes P

Prediction network

$H^{(0)}$

Encoder block × 5

LayerNorm

Temporal attention
4 heads

Geographic (c) attention
2 heads, M^G

Semantic attention
2 heads, M^s

Concat + Linear

residual

LayerNorm

Feed-forward network

residual

$H(t)$

Regime + precursor fusion (b)

Skip + order-progress fusion

Graph pooling + node evidence

Event heads
w, s, d, prefix

State / remaining time decoders

Process-cause classifier

Training loss

Event decoding
and report assembly

(c) Delay-conditioned interaction

Earlier precursor

Later constraint

lag

Pattern matching

Queue query patch

Activity prototypes P (b)

Q^P 38 × 20

K^P 16 × 20

V^P 16 × 20

$\alpha^P = \mathrm{softmax}(Q^P K^{PT} / \sqrt{d_h})$

Pattern weights 38 × 16

Pattern context C^P
38 × 20

Geographic key update

$\tilde{K}^G = K^G + C^P$

Base key K^G

Context C^P

Updated key $\tilde{K}^G$

Q^G, V^G

$\mathrm{softmax}(Q^G \tilde{K}^{GT} / \sqrt{d_h} + M^G)\, V^G$

Structured bottleneck report

schematic

Resource i*

t $t + \hat{s}$ $t + \hat{s} + \hat{d}$

State

Onset $\hat{s}$

Occurrence $\hat{w}$

Duration $\hat{d}$

Process cause $\hat{c}$

process-evidence class

Order remaining time $\hat{\ell}$

Minute-level prediction for the selected resource